Sequences and Series: Introduction

Questions 1–53

Question 1. For each of the following sequences, explain whether they can be described as convergent, divergent, periodic, or none of these.

a) $1, 2, 4, 8, 16, 32, \ldots$

b) $1, 5, 1, 5, 1, 5, \ldots$

c) $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \ldots$

d) $50, 60, 65, 67.5, 68.75, 69.375, \ldots$

Question 2. Sigma Notation

Evaluate each of these:

a) $\sum_{n=1}^{4}(5n-1)$

b) $\sum_{n=1}^{3}(3^n+n^2)$

c) $\sum_{n=1}^{5}\frac{2}{n}$

d) $\sum_{n=1}^{4}4\times\left(\frac{1}{2}\right)^{n-1}$

Question 3. Evaluate each of these:

a) $\sum_{n=3}^{6}(n^2 - 10)$

b) $\sum_{n=99}^{100}\frac{3}{n}$

c) $\sum_{n=6}^{8}50 \times 5^{n-5}$

Question 4. Simplify each of these, where k is a constant:

a) $\sum_{n=1}^{5}(n + k)$

b) $\sum_{n=1}^{4}(1 - kn^3)$

c) $\sum_{n=2}^{4}\frac{k}{n+1}$

d) $\sum_{n=3}^{6}\frac{kn}{k+1}$

Question 5. It is given that $f(n)$ is the nth term of a sequence. Some further information about $f(n)$ is given in the table below.

k	19	20	21	22	23	24	25
$\sum_{n=1}^{k} f(n)$	$\frac{19}{10}$	$\frac{21}{10}$	$\frac{58}{25}$	$\frac{127}{50}$	$\frac{69}{25}$	3	$\frac{163}{50}$

Find the value of

$$\sum_{n=20}^{24} f(n)$$

Question 6. It is given that

$$g(k) = \sum_{n=1}^{k} (n+2)^2$$

where k is an integer such that $k \geq 1$. Find:

a) $g(2)$
b) $g(5)$
c) $g(7) - g(6)$
d) $g(k+1) - g(k)$

Question 7. Find

$$\sum_{i=1}^{100} 1 + \sum_{i=20}^{100} 1 + \sum_{i=50}^{80} 1$$

Question 8. Show that

$$\sum_{n=1}^{4} \frac{6(n+1)}{\sqrt{n}} = 27 + 9\sqrt{2} + 8\sqrt{3}$$

Question 9. Show that

$$\sum_{n=1}^{3} \frac{n}{\sqrt{n}+\sqrt{n+1}} = 5 - \sqrt{2} - \sqrt{3}$$

Question 10. Show that

$$\sum_{n=1}^{3} \ln(n+1) = \ln 24$$

Question 11. Show that

$$\sum_{n=1}^{4} \ln\left(\frac{1}{n}\right) = -\ln 24$$

Question 12. Show that

$$\sum_{n=1}^{5} \ln\left(\frac{n+1}{n}\right) = \ln 6$$

Question 13. It is given that

$$h(k) = \sum_{n=1}^{k} \frac{n}{n+1}$$

Find the value of k such that

$$h(k+1) - h(k) = \frac{99}{100}$$

Question 14. Recurrence Relations and Limits

It is given that

$$u_{n+1} = 3u_n + 4$$

with $u_1 = 50$. Find the values of u_2, u_3 and u_4.

Question 15. It is given that

$$u_{n+1} = \frac{u_n + 2}{10}$$

with $u_1 = 1000$. Find the values of u_2, u_3 and u_4.

Question 16. It is given that

$$u_{n+1} = 5u_n + k$$

where k is a constant, with $u_1 = 2$ and $u_3 = 4$. Find the value of k.

Question 17. It is given that

$$u_{n+1} = \frac{1}{u_n + 2}$$

with $u_1 = 3$. Calculate

$$\sum_{n=1}^{3} u_n$$

Question 18. It is given that

$$u_{n+1} = \frac{10}{u_n} - k$$

where k is a constant, with $u_1 = 10$. It is also given that

$$\sum_{n=1}^{3} u_n = 0$$

Find the two possible values of k.

Question 19. In each of the following cases, use your calculator to determine if the sequence converges or diverges.

a) $u_{n+1} = 0.3u_n - 8,\ u_1 = 1$
b) $u_{n+1} = 1.1u_n - 3,\ u_1 = 2$
c) $u_{n+1} = 2.5u_n + 2,\ u_1 = -3$
d) $u_{n+1} = 3 - 1.8u_n,\ u_1 = 4$
e) $u_{n+1} = 100 - 0.9u_n,\ u_1 = 10$

Question 20. In each of the following cases, determine if the sequence converges or diverges.

a) $u_n = 3 + 0.6n$
b) $u_n = 2^n$
c) $u_n = \left(\frac{1}{3}\right)^n$
d) $u_n = \frac{10}{n+1}$

Question 21. In each of the following cases, it is given that the limit of u_n as n tends to infinity is L. In each case, write down an equation for L and find its value.

a) $u_{n+1} = \frac{u_n}{3} + 50$
b) $u_{n+1} = 0.85u_n - 6$
c) $u_{n+1} = \frac{u_n + 15}{30}$
d) $u_{n+1} = \frac{0.2u_n - 100}{25}$

Question 22. Arnold takes an initial dose of 400 mg of a certain type of painkiller, and every four hours must take another dose. The amount of painkiller in Arnold's system at the end of a four-hour period, just after he has taken the next dose, can be modelled by the recurrence relation $P_{n+1} = 0.25P_n + 400$.

a) Explain the significance of 0.25 in the recurrence relation.
b) Show that the amount of painkiller in Arnold's system 12 hours after he took the first dose, is 531 mg to the nearest mg.
c) Use algebra to determine the long-term amount of painkiller in Arnold's system.

Question 23. In each of the following cases, determine the possible value(s) of u_1 that will generate a constant sequence.

a) $u_{n+1} = \frac{9u_n}{8} + 5$
b) $u_{n+1} = \frac{10-u_n}{15}$
c) $u_{n+1} = \frac{5}{2} + \frac{21}{u_n}$
d) $u_{n+1} = \frac{55-3u_n-8u_n^2}{10u_n}$

Question 24. Write down the limit of the following sequences:

a) $u_n = \frac{1}{n+10}$
b) $u_n = \frac{n}{n+10}$
c) $u_n = \frac{n+11}{n+10}$
d) $u_n = \frac{2n+11}{n+10}$
e) $u_n = \frac{2n+11}{n^2+10}$

Question 25. It is given that $u_{n+1} = 1 - u_n$ with $u_1 = 1$. Find the value of u_{155}.

Question 26. It is given that $u_{n+1} = 8 - u_n$ with $u_1 = 1$. Find the value of u_{88}.

Question 27. It is given that $u_{n+1} = -u_n$ with $u_1 = 1$.

a) Find the value of u_{100}

b) Find the value of $\sum_{n=1}^{100} u_n$

Question 28. It is given that $u_{n+1} = 13 - u_n^2$ with $u_1 = 4$.

a) Find the value of u_{500}

b) Find the value of $\sum_{n=1}^{500} u_n$

Question 29. It is given that $u_{n+1} = -\frac{1}{u_n}$ with $u_1 = \frac{5}{3}$.

a) Find the value of u_{99}

b) Find the value of $\sum_{n=1}^{99} u_n$

Question 30. It is given that

$$u_{n+1} = \frac{u_n - 1}{k u_n}$$

with $u_1 = 3$, where k is a constant, generates a sequence with period 3.

a) Find the value of k

b) Hence find the value of $\sum_{n=1}^{100} u_n$

Question 31. It is given that $u_{n+3} = u_n$ with $u_1 = 4$, $u_2 = -6$ and $u_3 = 8$. Find the value of

$$\sum_{n=1}^{500} u_n$$

Question 32. It is given that $u_{n+4} = u_n$ with $u_1 = 1$, $u_2 = 4$, $u_3 = -7$ and $u_4 = -2$. Find the value of

$$\sum_{n=1}^{499} u_n$$

Question 33. It is given that $u_{n+3} = u_n + 2$ with $u_1 = 0$, $u_2 = 1$ and $u_3 = 5$. Find k such that $u_k = 200$.

Question 34. Periodic, Increasing, Decreasing
It is given that

$$u_n = 3 + (-1)^n$$

is a periodic sequence. State the period of the sequence.

Question 35. It is given that

$$u_n = \sin\left(\frac{3\pi n}{2}\right)$$

is a periodic sequence. State the period of the sequence.

Question 36. It is given that

$$u_n = \cos\left(\frac{2\pi n}{3}\right)$$

is a periodic sequence. State the period of the sequence.

Question 37. It is given that

$$u_n = \sin\left(\frac{2\pi n}{3}\right) - \cos\left(\frac{5\pi n}{3}\right)$$

is a periodic sequence. State the period of the sequence.

Question 38. It is given that

$$u_n = \cos\left(\frac{\pi n}{4}\right) - \sin\left(\frac{5\pi n}{2}\right)$$

is a periodic sequence. State the period of the sequence.

Question 39. It is given that

$$u_n = (-1)^n + \sin\left(\frac{\pi n}{2}\right)$$

is a periodic sequence. State the period of the sequence.

Question 40. It is given that

$$u_n = 3(-1)^n + \cos\left(\frac{\pi n}{3}\right)$$

is a periodic sequence. State the period of the sequence.

Question 41. It is given that

$$u_n = 2(-1)^n - \frac{1}{\cos(\pi n/3)}$$

is a periodic sequence. State the period of the sequence.

Question 42. Show that

$$u_n = \frac{3n}{2+n^2}$$

is not an increasing sequence.

Question 43. Show that

$$u_n = \frac{n+1}{n^3+n-1}$$

is not an increasing sequence.

Question 44. Show that

$$u_n = n - \frac{1}{n+1}$$

is not a decreasing sequence.

Question 45. Show that

$$u_n = n^2 - 100n$$

is not a decreasing sequence.

Question 46. Prove that

$$u_n = 3^{-n}$$

is a decreasing sequence.

Question 47. Prove that

$$u_n = \frac{1}{n^2}$$

is a decreasing sequence.

Question 48. Prove that

$$u_n = \frac{n+1}{n+2}$$

is an increasing sequence.

Question 49. Prove that

$$u_n = \frac{n+3}{n+2}$$

is a decreasing sequence.

Question 50. Prove that

$$u_n = \frac{5}{n+5}$$

is a decreasing sequence.

Question 51. Prove that

$$u_n = \frac{2-n}{n}$$

is a decreasing sequence.

Question 52. Prove that

$$u_n = (2n+1)^2 - (n-1)^2$$

is an increasing sequence.

Question 53. Prove that

$$u_n = \sqrt{n+1} + \sqrt{n}$$

is an increasing sequence.